Name
Date
Course

# Beginning Algebra Final Test (Part II)

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
iii) Draw a rectangle around your answers.
iv) Staple a copy of these test questions with your work and your final answers.

1 Find an equation of the line through the following two points in

$(-1,3)$ & $(-4,1)$

a) Point-Slope form. b) Slope-intercept form.

c) What is the slope of the line?

d) What is the y-intercept?

2 Use the slope-intercept form to find the slope-intercept form of the equation with slope $-3$ and point $(-4,-3)$ on the line.

3 Find an equation of the line with the following characteristics.

X-intercept: 8
slope: $-1$

4 Find an equation of the line with y-intercept 2 and that is perpendicular to the line with equation $4x+10=y$.

5 Find an equation of the line through the point $(-4,5)$ and parallel to the line with equation $y+5=-\frac{3}{7}(x+2)$.

6 Find an equation of the line with y-intercept $-3$ and that is perpendicular to the line with equation $x-y=-2$.

7 Convert the following equation to slope-intercept form.

$$-2(x-2)=y-7$$

Solve.

8 $2\sqrt{x}=5\sqrt{3x-1}$

9 $-2=1-\dfrac{2\sqrt{\frac{x}{5}+1}}{3}$

10 a) $\sqrt{x^2-8x+23}=-4$ b) $\sqrt{3x+1}-\sqrt{x+8}=1$

11 $3=4+y^2$

12 $9(x+1)^2-5=7$

13 a) Solve the following system by the graphical method. Use graphing paper.

$$\begin{cases} y=2x \\ 4x-y=2 \end{cases}$$

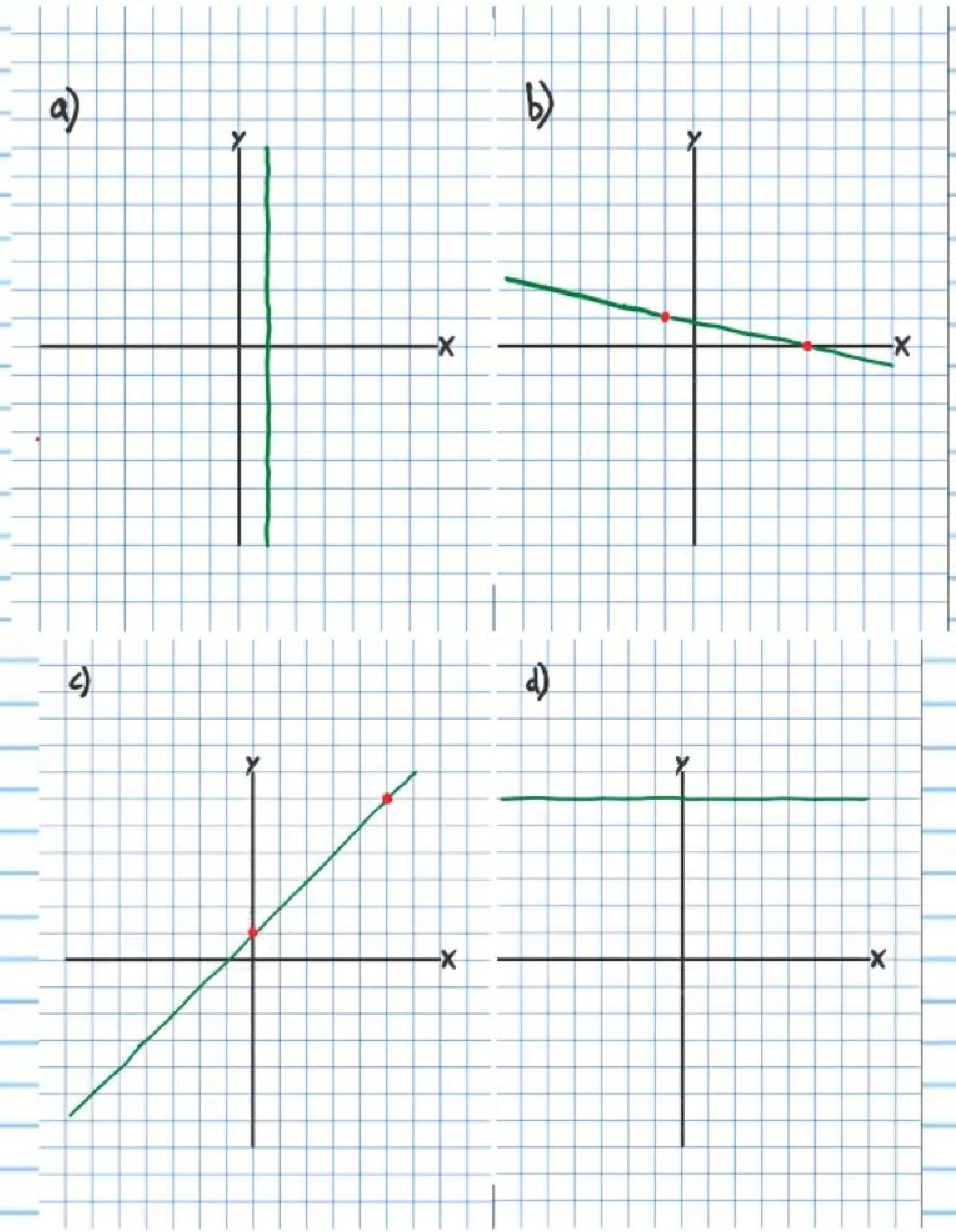
b) Then solve the system algebraically (i.e. substitution or elimination) to check your solution.

Solve.

14 $x-\dfrac{14}{5}=-\dfrac{8}{5x}$

15 $\dfrac{9x}{3x-3x^3}+\dfrac{-6}{x-1}=\dfrac{1}{x+1}$

(16) For each graph below, write an equation of the line shown and identify the slope.




(17) Graph the following equations and identify the slopes. Label axes and intervals.

a) $y=-3$ b) $x=4$

Solve.

(18) $\frac{4}{x-3}-\frac{3}{x+5}=\frac{x^2-43}{x^2+2x-15}$

(19) $2+\frac{4}{x-7}=\frac{-18}{x+3}$

(20) There are balloons and party hats in a bag. The number of party hats is five less than twice the number of balloons. The difference between the number of party hats and the number of balloons is seven. How many party hats are in the bag?

(21) Angelina bought four pounds of almonds and five pounds of walnuts for \$43. Bill bought three pounds of almonds and four pounds of walnuts for \$33. How much does each item cost per pound?

(22) There are only quarters and dimes on a table. There are 36 coins on the table. The total value of these coins is \$5.85. How many of each type of coin are on the table?

Solve.

(23) $\begin{cases} -4x+3y=2 \\ x=-2y+3 \end{cases}$

(24) A boy throws a baseball into the air. The height of the ball (in feet) $t$ seconds after release is given by the equation below.

$$h = -16t^2 + 12t + 4$$

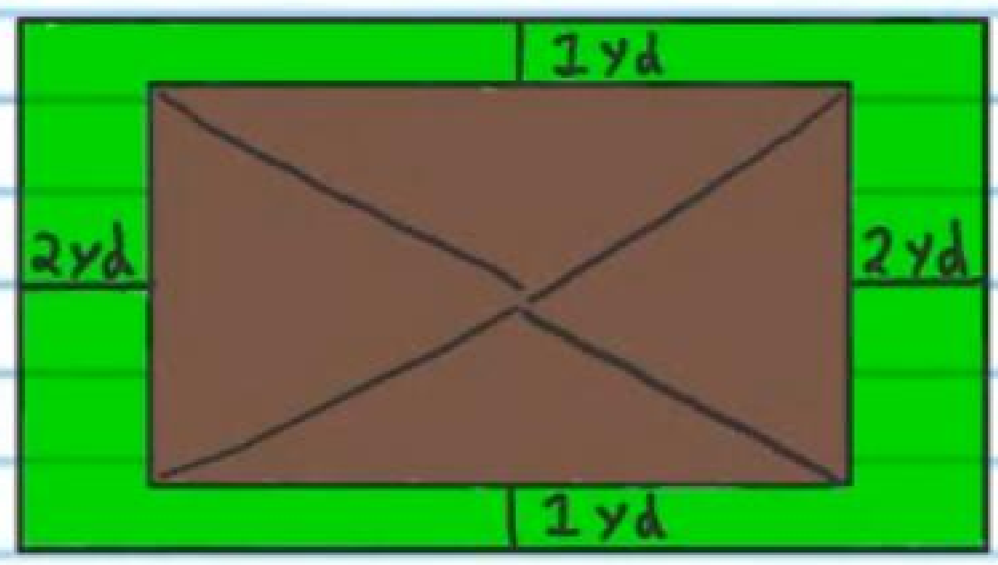
a) What is the height of the ball $\frac{2}{3}$ of a second after release?

b) How many seconds after release will the ball be six feet in the air (Hint: There are two answers)?

c) How many seconds after release will the ball be four feet off the ground (Hint: There are two answers)?

d) How many seconds after release will it take for the ball to hit the ground (Hint: There will be two answers, one that makes sense and another that is extraneous)?

(25) Thomas has a garden that surrounds his house. The area of the land that his house is on is 50 square yards less than the area of the garden and land his house is on combined. The length of his house is 3 yards less than the width. What are the dimensions of his house?




☆ Picture not drawn to scale.

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# Beginning Algebra Final Test (Part II) Answers

① a) $\boxed{y-3=\frac{2}{3}(x+1)}$ or $\boxed{y-1=\frac{2}{3}(x+4)}$ b) $\boxed{y=\frac{2}{3}x+\frac{11}{3}}$

c) $\boxed{m=\frac{2}{3}}$ d) $\boxed{b=\frac{11}{3} \text{ or } 3\frac{2}{3}}$

② $\boxed{y=-3x-15}$

③ $\boxed{y-0=-(x-8)}$ or $\boxed{y=-x+8}$

④ $\boxed{y=-\frac{1}{4}x+2}$ or $\boxed{y=-\frac{x}{4}+2}$

⑤ $\boxed{y-5=-\frac{3}{7}(x+4)}$

⑥ $\boxed{y=-x-3}$

⑦ $\boxed{y=-2x+11}$

⑧ $\boxed{x=\frac{25}{71}}$

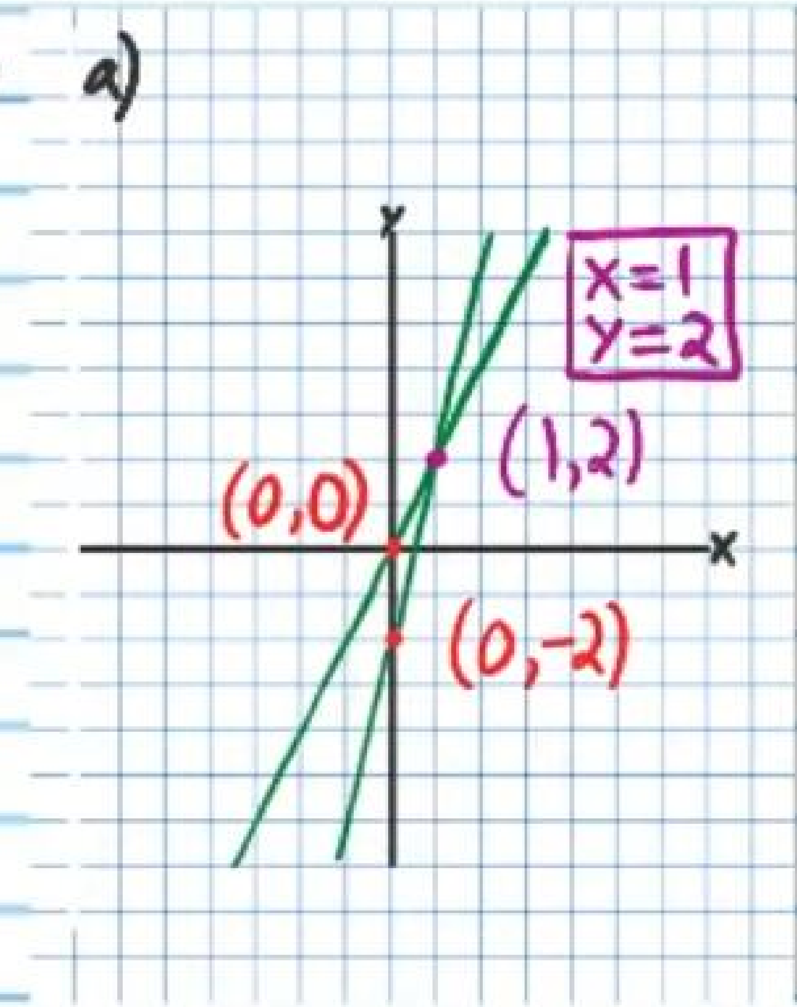
⑨ $\boxed{x=\frac{385}{4} \text{ or } 96\frac{1}{4}}$

⑩ a) $\boxed{\text{No Solution}}$ b) $\boxed{x=8}$ $x\neq 1$

⑪ $\boxed{\text{No Solution}}$

⑫ $\boxed{x=\pm\frac{2\sqrt{3}}{3}-1}$

⑬ a)




b) Full credit if solution produces $x=1$ and $y=2$ and work is neat and correct.

⑭ $\boxed{x=\frac{4}{5}}$, $\boxed{x=2}$

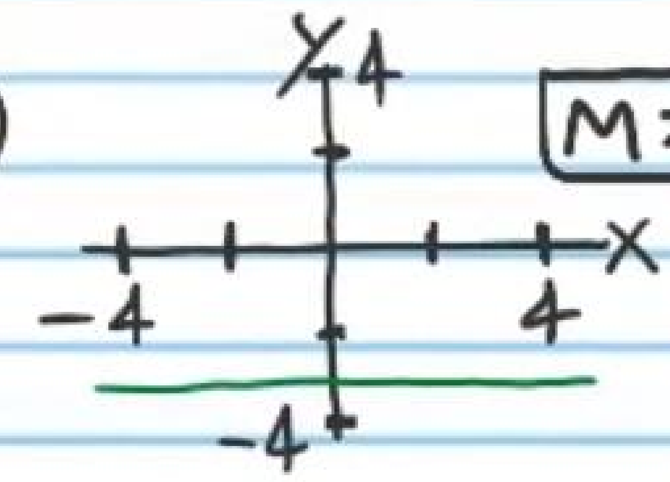
⑮ $\boxed{x=-\frac{8}{7}}$ $x\neq 0$

⑯ a) $\boxed{x=1}$, $\boxed{m=\text{undefined}}$

b) $\boxed{y-1=-\frac{1}{5}(x+1)}$ or $\boxed{y-0=-\frac{1}{5}(x-4)}$, $\boxed{m=-\frac{1}{5}}$

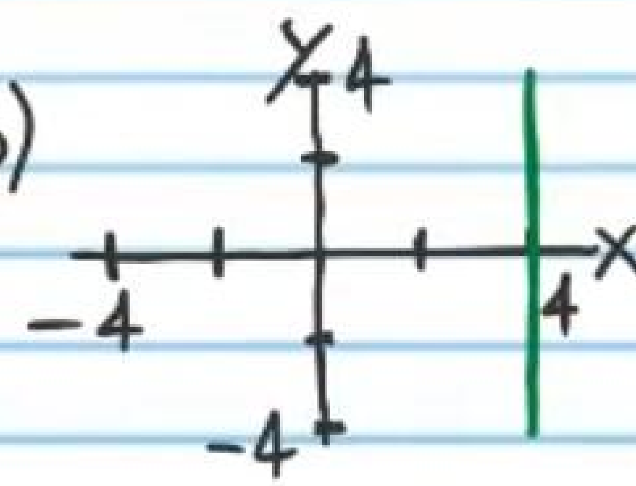
c) $\boxed{y=x+1}$ $\boxed{m=1}$ d) $\boxed{y=6}$ $\boxed{m=0}$

⑰ a)




$\boxed{m=0}$

b)




$\boxed{m=\text{und}}$

⑱ $\boxed{x=9}$, $\boxed{x=-8}$

(19) $\boxed{x=6}$, $\boxed{x=-13}$

(20) $\boxed{P=19}$

(21) $\boxed{A=\$7/\text{lbs}}$ $\boxed{W=\$3/\text{lbs}}$

(22) $\boxed{Q=15}$, $\boxed{D=21}$

(23) $\boxed{x=5/11}$, $\boxed{y=14/11}$

(24) a) $\boxed{\frac{44}{9}\text{ ft}}$ or $\boxed{4\frac{8}{9}\text{ ft}}$ b) $\boxed{t=1/4\text{ seconds}}$, $\boxed{t=\frac{1}{2}\text{ seconds}}$

c) $\boxed{t=0\text{ seconds}}$, $\boxed{t=3/4\text{ of a second}}$

d) $\boxed{t=1\text{ second}}$, $t\neq-\frac{1}{4}$ seconds

(25) $\boxed{W=9\text{ yd}}$ $\boxed{L=6\text{ yd}}$